

Lab 6 - Factorial Designs

ENVX2001 Applied Statistical Methods

Semester 1, 2026

💡 Learning outcomes

At the end of this practical students should be able to:

- use R to analyse experiments with a factorial treatment structure where the experimental design is a CRD or RCBD;

All of the data for this practical is in the **Data6.xlsx** file.

Exercise 1 - More Treatment Structures (Walk-through)

A microbiologist conducted an experiment to assess the survival of *Salmonella typhinerium* when subjected to various treatments. A factorial treatment design was used, the treatments being various combinations of sorbic acid, pH, and water activity. The density of Salmonella seven days after the treatment began was recorded. The Salmonella data is in the form:

$$\log_e(density/ml)$$

, to 2 decimal places.

The data is found in the **Salmonella** sheet in the **Data6.xlsx** file.

There is a slight **trick** to analysing this experiment which will improve your ability to analyse factorial experiments in general.

The table below shows you how to specify the treatment structure and individual interaction term in a factorial design involving 2 factors, A and B.

term	command
2-way factorial	A*B
2-way interaction	A:B

Question 1.1

(i) Import the data into R and describe the data using numerical and graphical summaries. In particular does the exploratory data analysis (EDA) show any difference in *S.typhinerium* density between the levels of each of the treatment factors (a) sorbic acid (b) pH (c) water activity? Is there evidence of an interaction using interaction plots. No need to look at marginal means as the interaction plots summarise these.

Tip

Some hints for the EDA are:

- in many experiments we will have treatment factors which have a numerical value (as is the case here) but for ANOVA-type problems we are comparing the mean response between each value so we need to tell R the data type is a factor. If we don't, it will fit a regression model which is Topics 7-9. This will result in the wrong type of analysis.
- use the `tapply` function in conjunction with the `summary` function;
- boxplots for individual levels of a factor are likely to be the most informative as compared to numerical summaries they visually show the spread of the observations as well. Example R code is below.

```
CODE
boxplot(Density~pH,data=salm)
```

- to create interaction plots use the `interaction.plot` or `emmip` function. You will need one of these for each 2-way interaction in your model.

```
CODE
#
```

Solution

```
CODE
library(readxl)
salm<-read_excel("data/Data6.xlsx",sheet="Salmoneλλα")
str(salm)
```

```
OUTPUT
tibble [45 × 4] (S3: tbl_df/tbl/data.frame)
 $ Sorbic.Acid: num [1:45] 0 0 0 100 100 100 200 200 200 0 ...
 $ pH         : num [1:45] 5 5.5 6 5 5.5 6 5 5.5 6 5 ...
 $ Activity   : num [1:45] 0.78 0.78 0.78 0.78 0.78 0.78 0.78 0.78 0.78 0.82 ...
 $ Density    : num [1:45] 4.2 4.34 4.31 4.18 4.39 4.13 4.15 4.12 3.93 4.52 ...
```

CODE

```
salm$Sorbic.Acid<-as.factor(salm$Sorbic.Acid)
salm$pH<-as.factor(salm$pH)
salm$Activity<-as.factor(salm$Activity)
str(salm)
```

OUTPUT

```
tibble [45 × 4] (S3: tbl_df/tbl/data.frame)
 $ Sorbic.Acid: Factor w/ 3 levels "0","100","200": 1 1 1 2 2 2 3 3 3 1 ...
 $ pH         : Factor w/ 3 levels "5","5.5","6": 1 2 3 1 2 3 1 2 3 1 ...
 $ Activity   : Factor w/ 5 levels "0.78","0.82",..: 1 1 1 1 1 1 1 1 2 ...
 $ Density    : num [1:45] 4.2 4.34 4.31 4.18 4.39 4.13 4.15 4.12 3.93 4.52 ...
```

First we calculate summary statistics for different levels of each treatment factor.

CODE

```
tapply(salm$Density,salm$Sorbic.Acid,summary)
```

OUTPUT

```
$`0`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 4.200  4.430   5.060   5.303   6.060   6.700

$`100`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 4.130  4.290   4.850   4.999   5.530   6.520

$`200`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 3.930  4.265   4.410   4.891   5.315   6.430
```

CODE

```
tapply(salm$Density,salm$pH,summary)
```

OUTPUT

```
$`5`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 4.150  4.245   4.790   5.082   5.960   6.510

$`5.5`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 4.120  4.365   4.950   5.066   5.665   6.700

$`6`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 3.930  4.300   4.850   5.045   5.535   6.650
```

CODE

```
tapply(salm$Density,salm$Activity,summary)
```

OUTPUT

```
$`0.78`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 3.930  4.130   4.180   4.194   4.310   4.390
```

```

$`0.82`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 4.180  4.270  4.310  4.387  4.430  4.850

$`0.86`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 4.29   4.41   4.85   4.79   5.01   5.35

$`0.9`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 5.010  5.200  5.430  5.532  5.870  6.140

$`0.94`
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 6.180  6.250  6.430  6.418  6.520  6.700

```

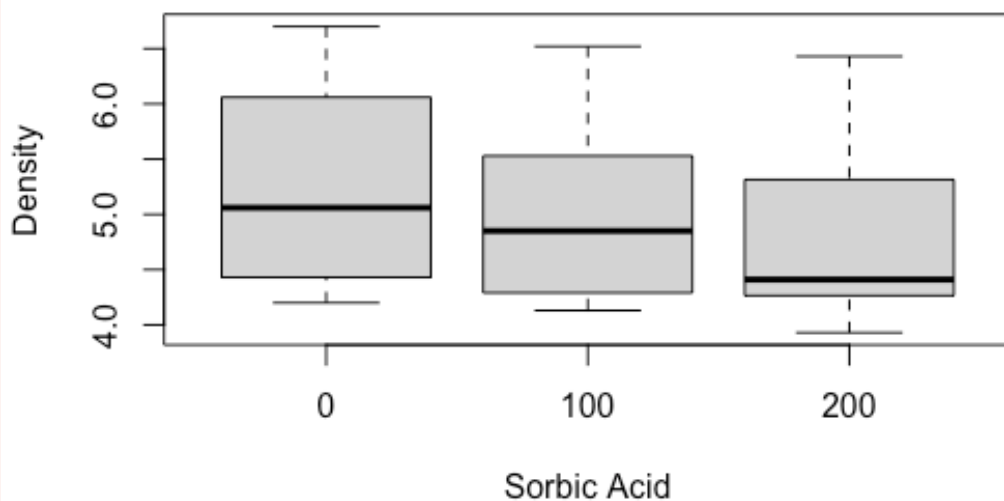
There are a lot of statistics to examine which is where graphical summaries are easier to interpret.

The boxplots for each level of Sorbic.Acid show that Density decreases as the Sorbic.Acid increases, this is based off the median values. However it should be noted the distributions at each level do overlap each other which may indicate no significant differences.

```

CODE
boxplot(Density~Sorbic.Acid,data=salm,main=" ",xlab="Sorbic Acid",ylab="Density")

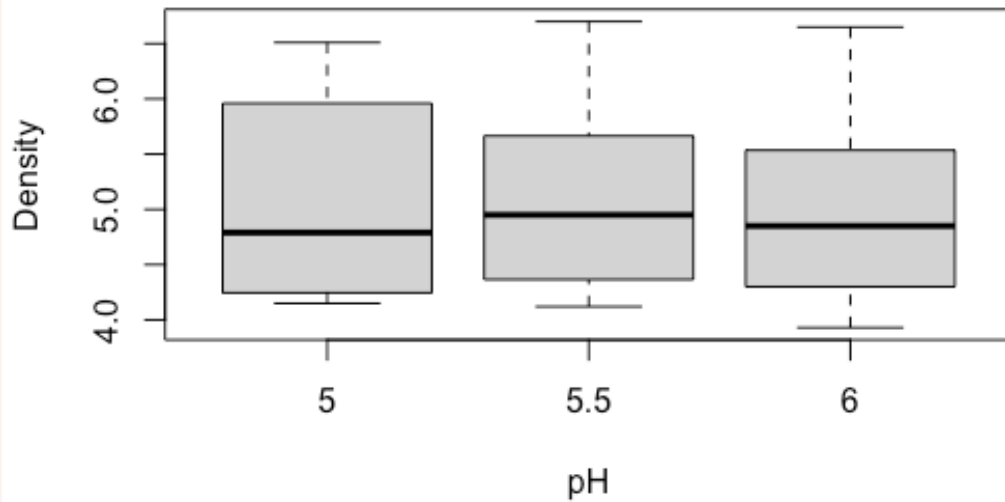
```



The boxplots for each level of pH show that there is little difference in Density, this is based off the median values.

CODE

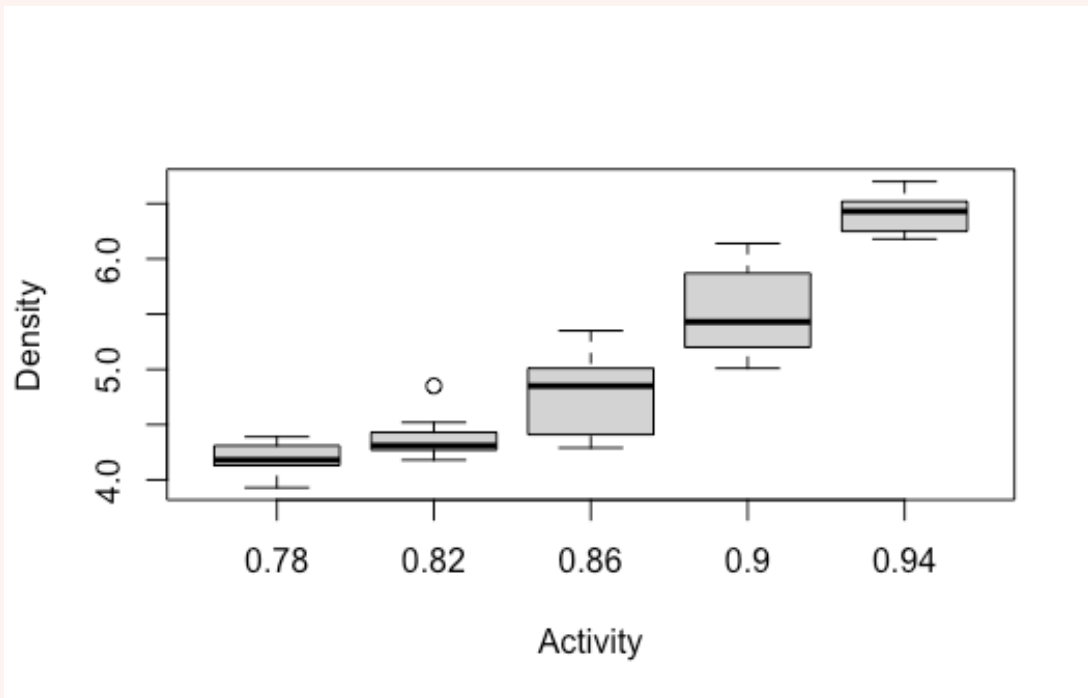
```
boxplot(Density~pH,data=salm,main=" ",xlab="pH",ylab="Density")
```



The boxplots for each level of Activity show that Density increases as the Activity increases quite markedly, this is based off the median values. Also note the distributions for each level in some cases do not overlap at all, for example 0.78 and 0.82 are quite distinct from 0.9 and 0.94.

CODE

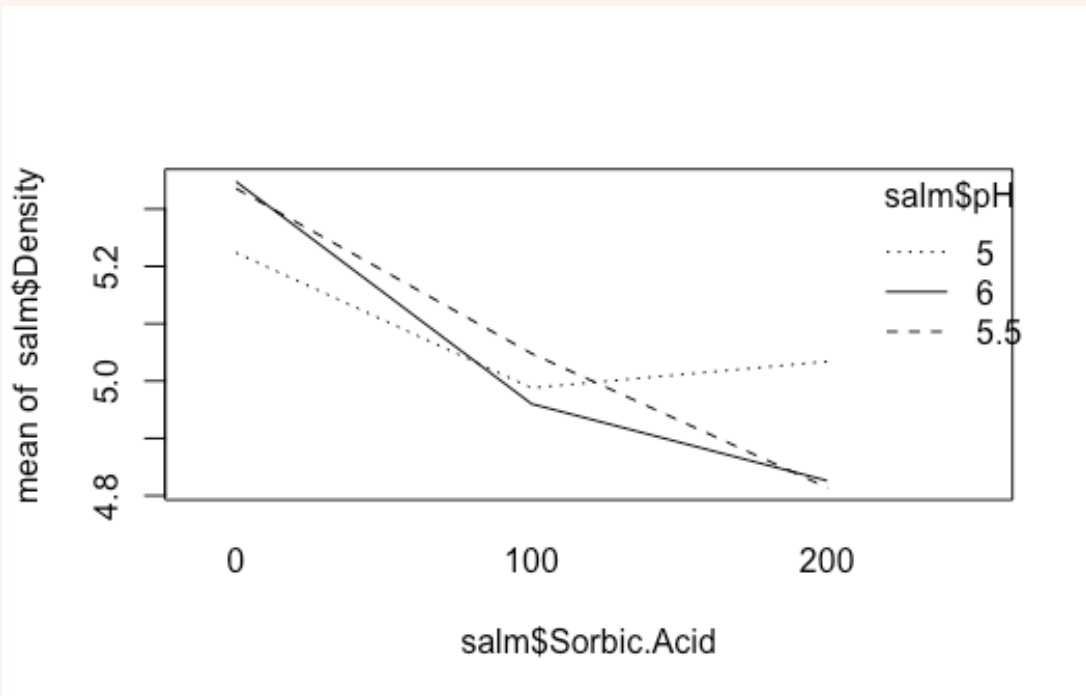
```
boxplot(Density~Activity,data=salm,main=" ",xlab="Activity",ylab="Density")
```



There are three 2 factor interactions which we can explore with interaction plots. When interpreting the interaction plots we are looking for changes in the shape of the lines for different levels of the treatment factor shown in the inset.

CODE

```
interaction.plot(salm$Sorbic.Acid, salm$pH, salm$Density)
```



Or:

CODE

```
library(emmeans)
```

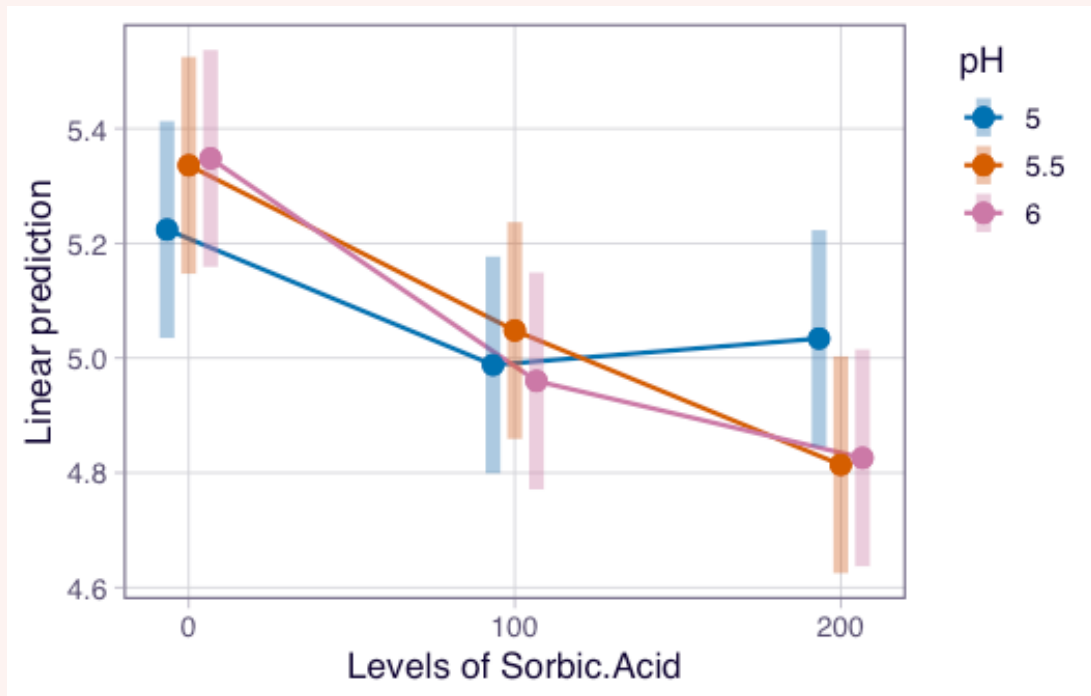
OUTPUT

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

CODE

```
salm.aov.2<-aov(Density~Sorbic.Acidity*pH*Activity-Sorbic.Acidity:pH:Activity,data=salm)

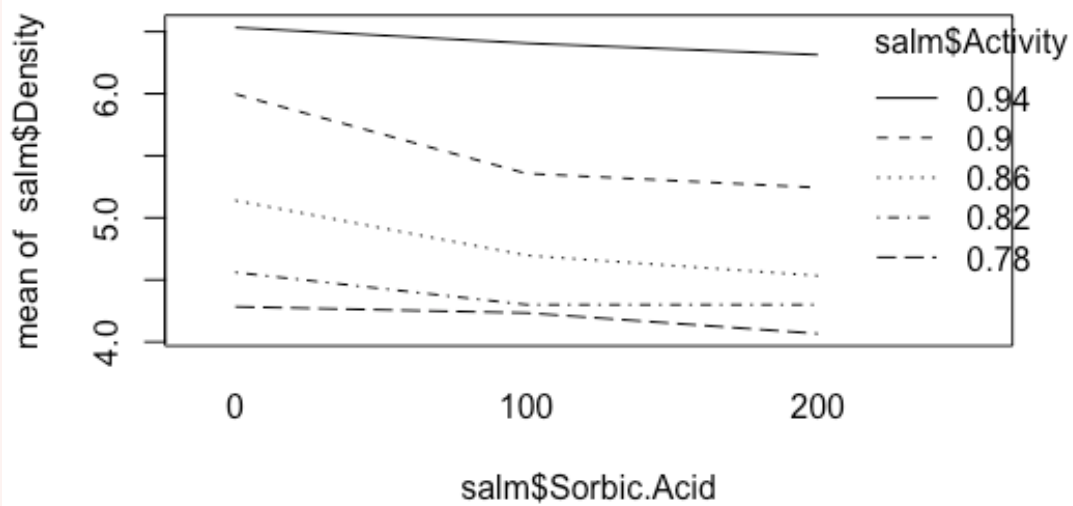
emmip(salm.aov.2, pH ~ Sorbic.Acidity, CIs = TRUE)
```



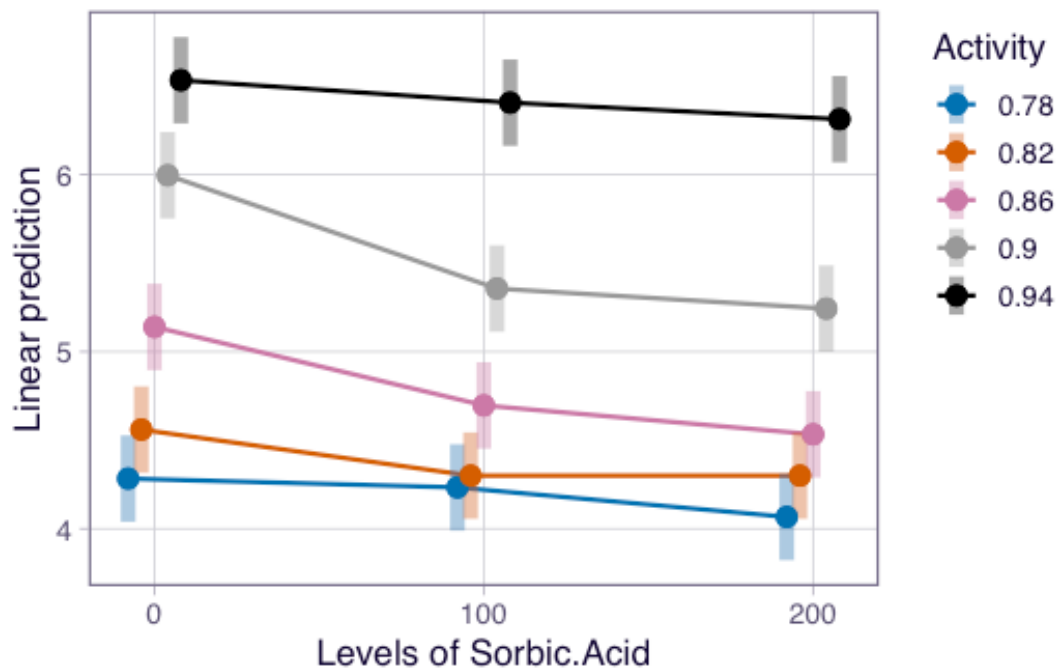
The pH and Sorbic.Acid interaction plot does not have parallel lines, the difference being when pH = 5. There may be an interaction here but formal hypothesis testing is needed to determine this.

CODE

```
interaction.plot(salm$Sorbic.Acid,salm$Activity,salm$Density)
```

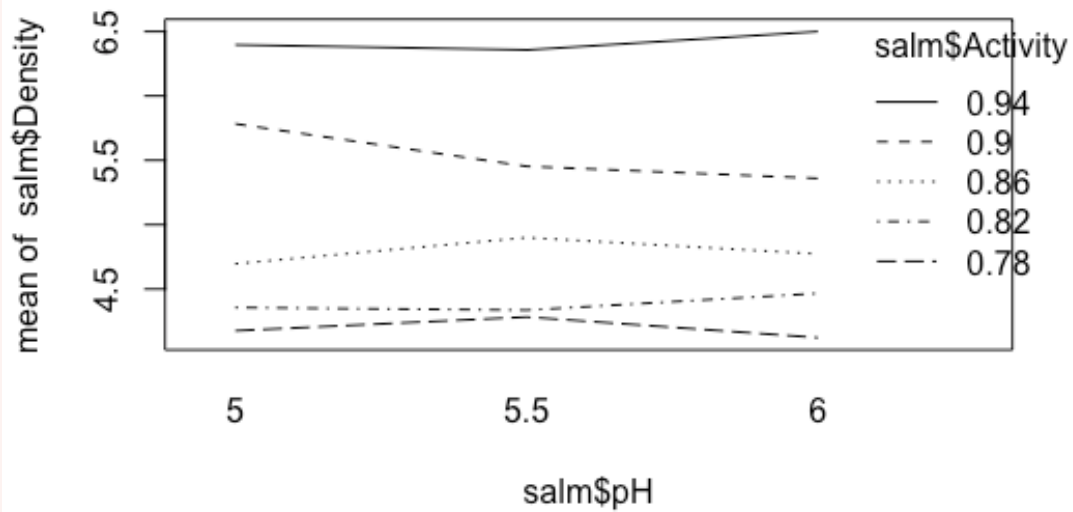
```
CODE
#Or
emmip(salm.aov.2, Activity ~ Sorbic.Acidity, CIs = TRUE)
```



The Activity and Sorbic.Acid interaction plot does have parallel lines so there is not likely to be an interaction. Note that the traces for different Activity levels are quite well separated in most cases which is further evidence of a treatment effect for Activity.

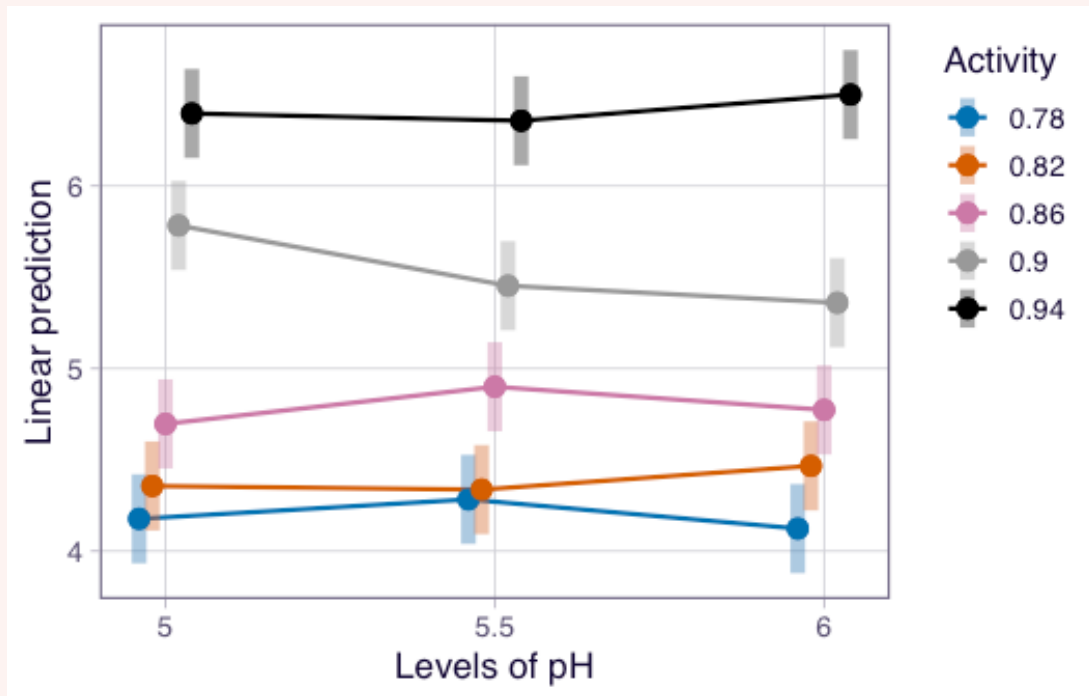
CODE

```
interaction.plot(salm$pH, salm$Activity, salm$Density)
```



CODE

```
#Or  
emmip(salm.aov.2, Activity ~ pH, CIs = TRUE)
```



The Activity and pH interaction plot does have parallel lines so there is not likely to be an interaction. Note that the traces for different Activity levels are quite well separated in most cases which is further evidence of a treatment effect for Activity.

Question 1.2

(ii) Write out the model you are fitting in terms of main effects and interactions, words are fine. Note, that each of these will have an associated statistical hypotheses. Obtain an ANOVA for these data, including appropriate interaction terms in the model.

What do you notice in the ANOVA table when you run the full 3-way factorial ANOVA? Use the summary function to extract the ANOVA table.

Run the model again but only include the interactions involving two factors: **do not include the three factor interaction term**. Why are we not including the three factor interaction?

💡 Tip

Look at the data in Excel and **manually** calculate the mean Density for when pH = 5.0, Activity = 0.78 and Sorbic.Acid = 0. How many observations were used to calculate the mean?

The model in words is:

CODE

```
#
```

Solution

density = sorbic acid + pH + activity (main effects) +

sorbic acid:pH + sorbic acid:activity + activity:pH (2 factor interactions) +

sorbic acid:pH:activity (3 factor interaction)

We are not fitting the 3 factor interaction as there is no replication at that level. For every combination of the three factors (Sorbic.Acid, Activity and pH), there is only one observation, so if this is fitted there will be zero degrees of freedom; replication would be needed to fit a three-way interaction. Effectively, by not fitting a three factor interaction, this term, becomes the Residual term.

To illustrate compare the output from a model with and without the 3 factor interaction.

CODE

```
salm.aov<-aov(Density~Sorbic.Acid*pH*Activity,data=salm)
summary(salm.aov)
```

OUTPUT

	Df	Sum Sq	Mean Sq
Sorbic.Acid	2	1.366	0.683
pH	2	0.011	0.005
Activity	4	30.077	7.519
Sorbic.Acid:pH	4	0.209	0.052
Sorbic.Acid:Activity	8	0.501	0.063
pH:Activity	8	0.452	0.056
Sorbic.Acid:pH:Activity	16	0.635	0.040

CODE

```
salm.aov.2<-aov(Density~Sorbic.Acid*pH*Activity-Sorbic.Acid:pH:Activity,data=salm)
summary(salm.aov.2)
```

OUTPUT

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Sorbic.Acid	2	1.366	0.683	17.196	0.000103 ***
pH	2	0.011	0.005	0.133	0.876834
Activity	4	30.077	7.519	189.367	2.96e-13 ***
Sorbic.Acid:pH	4	0.209	0.052	1.319	0.305270
Sorbic.Acid:Activity	8	0.501	0.063	1.578	0.208120
pH:Activity	8	0.452	0.056	1.422	0.260864
Residuals	16	0.635	0.040		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Question 1.3

(iii) Test that the model assumptions have been met.

CODE

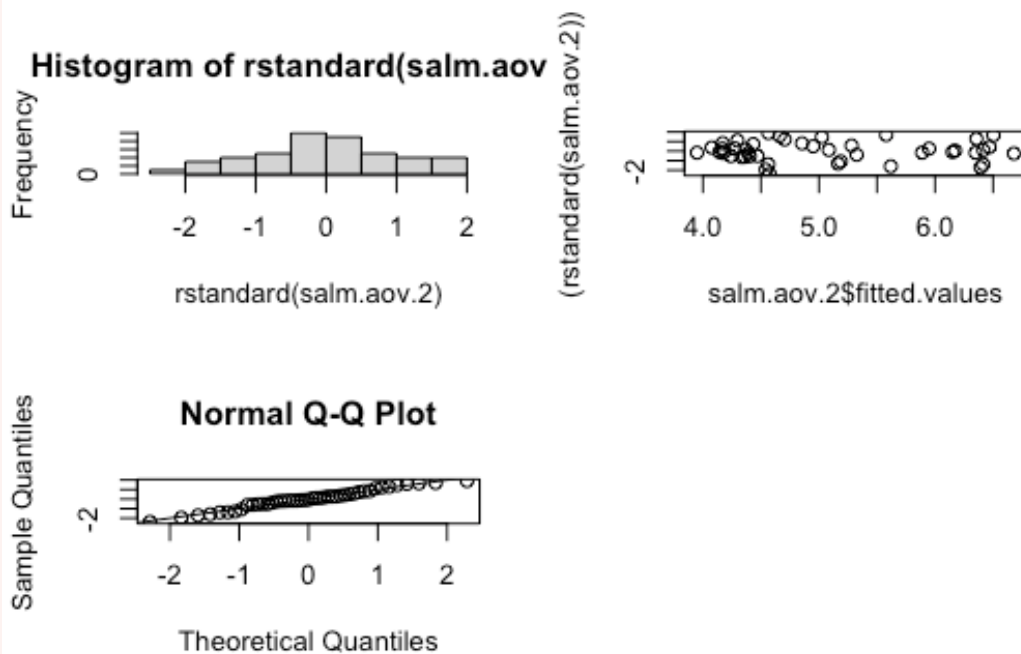
```
#
```

Solution

The residual diagnostics show that the data is normally distributed, has constant variance and has no obvious outliers. The response has been log transformed so this is no surprise.

CODE

```
par(mfrow=c(2,2))
hist(rstandard(salm.aov.2))
plot(salm.aov.2$fitted.values, (rstandard(salm.aov.2)))
qqnorm(rstandard(salm.aov.2))
abline(0,1)
```



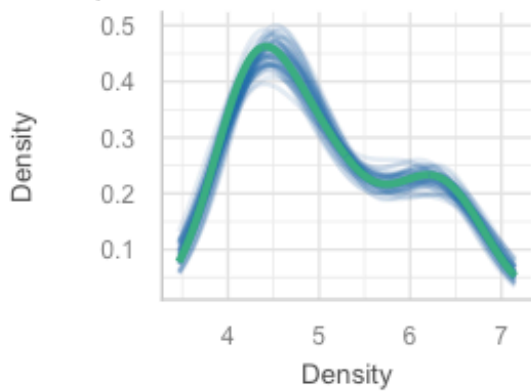
or

CODE

```
library(performance)
check_model(salm.aov.2)
```

Posterior Predictive Check

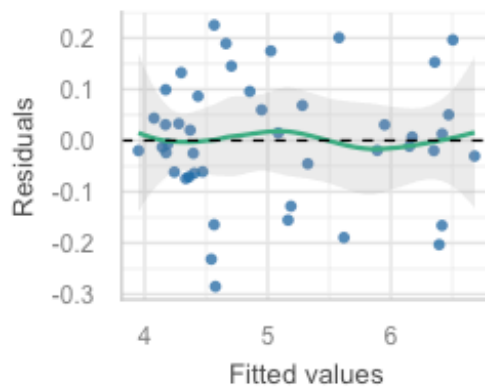
Model-predicted lines should resemble observed



— Observed data — Model-predicted data

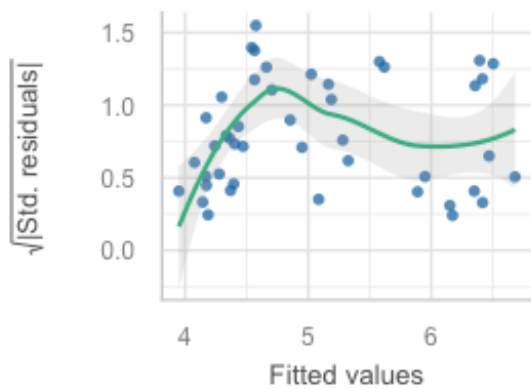
Linearity

Reference line should be flat and horizontal



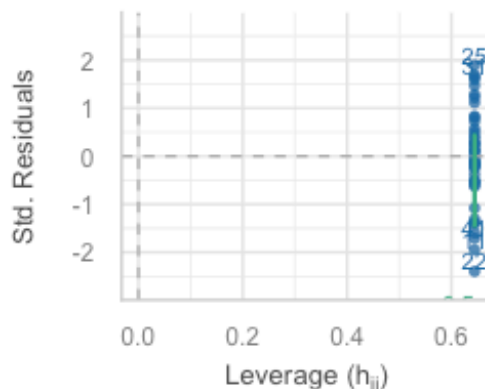
Homogeneity of Variance

Reference line should be flat and horizontal



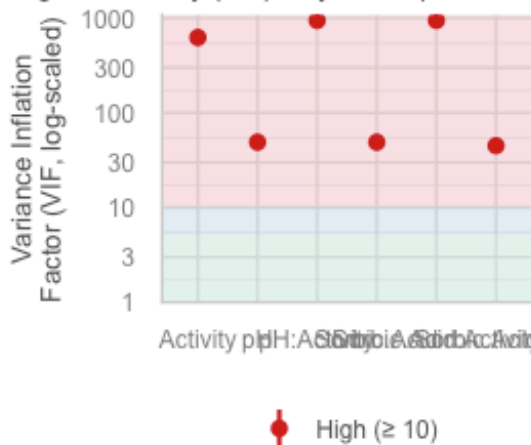
Influential Observations

Points should be inside the contour lines



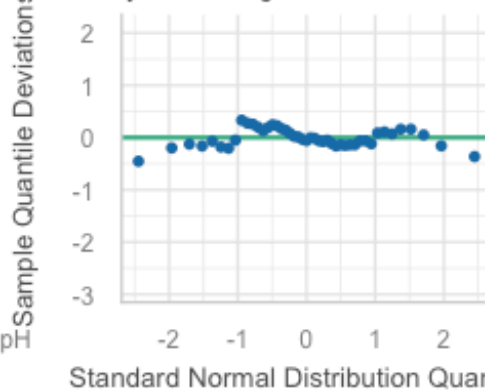
Collinearity

High collinearity (VIF) may inflate parameter



Normality of Residuals

Points should fall along the line



Question 1.4

(iv) What are the significant effects in the model? For the significant effects we conduct post-hoc tests the Tukey's Test to determine which pairs are significantly different. Write overall conclusions, in terms of the statistical hypothesis testing and in terms of the biological description of the experiment.

```
CODE
#
```

Solution

The ANOVA shows that none of the 2 factor interactions are significant which means we can examine the main effects individually. The significant effects are pH and Sorbic.Acid with P-values less than 0.05.

```
CODE
summary(salm.aov.2)
```

```
OUTPUT

              Df Sum Sq Mean Sq F value    Pr(>F)
Sorbic.Acid      2  1.366    0.683   17.196 0.000103 ***
pH                2   0.011    0.005    0.133 0.876834
Activity          4 30.077    7.519  189.367 2.96e-13 ***
Sorbic.Acid:pH    4   0.209    0.052    1.319 0.305270
Sorbic.Acid:Activity 8   0.501    0.063    1.578 0.208120
pH:Activity       8   0.452    0.056    1.422 0.260864
Residuals       16   0.635    0.040
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Tukey test for the significant effects is shown below.

```
CODE
emmeans(salm.aov.2, pairwise ~ Sorbic.Acid)
```

```
OUTPUT
NOTE: Results may be misleading due to involvement in interactions
```

```
OUTPUT
$emmeans
Sorbic.Acid emmean      SE df lower.CL upper.CL
0            5.30 0.0515 16     5.19     5.41
100          5.00 0.0515 16     4.89     5.11
200          4.89 0.0515 16     4.78     5.00

Results are averaged over the levels of: pH, Activity
Confidence level used: 0.95

$constrasts
contrast              estimate      SE df t.ratio p.value
```

Sorbic.Acids0 - Sorbic.Acids100	0.304	0.0728	16	4.178	0.0019
Sorbic.Acids0 - Sorbic.Acids200	0.411	0.0728	16	5.653	0.0001
Sorbic.Acids100 - Sorbic.Acids200	0.107	0.0728	16	1.475	0.3285

Results are averaged over the levels of: pH, Activity
P value adjustment: tukey method for comparing a family of 3 estimates

Increased concentrations of sorbic acid result in significant reductions in Salmonella levels. There was a far greater reduction in Salmonella when the the Sorbic.Acids was increased from 0 to 100, compared with 100 to 200, suggesting that a saturation effect may be occurring.

CODE

```
emmeans(salm.aov.2, pairwise ~ Activity)
```

OUTPUT

NOTE: Results may be misleading due to involvement in interactions

OUTPUT

\$emmeans

Activity	emmean	SE	df	lower.CL	upper.CL
0.78	4.19	0.0664	16	4.05	4.34
0.82	4.39	0.0664	16	4.25	4.53
0.86	4.79	0.0664	16	4.65	4.93
0.9	5.53	0.0664	16	5.39	5.67
0.94	6.42	0.0664	16	6.28	6.56

Results are averaged over the levels of: Sorbic.Acids, pH
Confidence level used: 0.95

\$contrasts

contrast	estimate	SE	df	t.ratio	p.value
Activity0.78 - Activity0.82	-0.192	0.0939	16	-2.046	0.2894
Activity0.78 - Activity0.86	-0.596	0.0939	16	-6.340	0.0001
Activity0.78 - Activity0.9	-1.338	0.0939	16	-14.241	<.0001
Activity0.78 - Activity0.94	-2.223	0.0939	16	-23.669	<.0001
Activity0.82 - Activity0.86	-0.403	0.0939	16	-4.294	0.0044
Activity0.82 - Activity0.9	-1.146	0.0939	16	-12.195	<.0001
Activity0.82 - Activity0.94	-2.031	0.0939	16	-21.622	<.0001
Activity0.86 - Activity0.9	-0.742	0.0939	16	-7.901	<.0001
Activity0.86 - Activity0.94	-1.628	0.0939	16	-17.329	<.0001
Activity0.9 - Activity0.94	-0.886	0.0939	16	-9.427	<.0001

Results are averaged over the levels of: Sorbic.Acids, pH
P value adjustment: tukey method for comparing a family of 5 estimates

As the Activity increases, the $\log_e$ (Density) of Salmonella increases approximately exponentially, and based on the Tukey Test the increase is significant at each increase in water activity with the exception of between 0.78 and 0.82.

Exercise 2 - More practice

An experiment was performed to study the control of potato blight on potatoes. A factorial treatment structure was employed with 3 Varieties in combination with 5 Chemical treatments. The experiment was conducted using a randomised complete block design (RCBD) with three blocks. The response is yield of potatoes (lbs).

Provide graphical summaries of the data, analyse the experiment, conduct any post-hoc tests if the results are significant. The data is found in the **Potato** sheet in the **Data6.xlsx** file.

```
CODE
#
```

Solution

First we read in the data and convert to factor if required.

```
CODE
library(readxl)
pot<-read_excel("data/Data6.xlsx",sheet="Potato")
str(pot)
```

```
OUTPUT
tibble [45 × 4] (S3: tbl_df/tbl/data.frame)
 $ Yield   : num [1:45] 34.1 76.3 72.2 51.9 42.9 62.1 66.3 39.1 75.1 41.7 ...
 $ Block   : num [1:45] 1 1 1 2 2 2 3 3 3 1 ...
 $ Variety : chr [1:45] "Pontiac" "Sebago" "Sebago" "Desire" ...
 $ Chemical: chr [1:45] "CuOxy4" "Dithane10" "Dithane4" "CuOxy4" ...
```

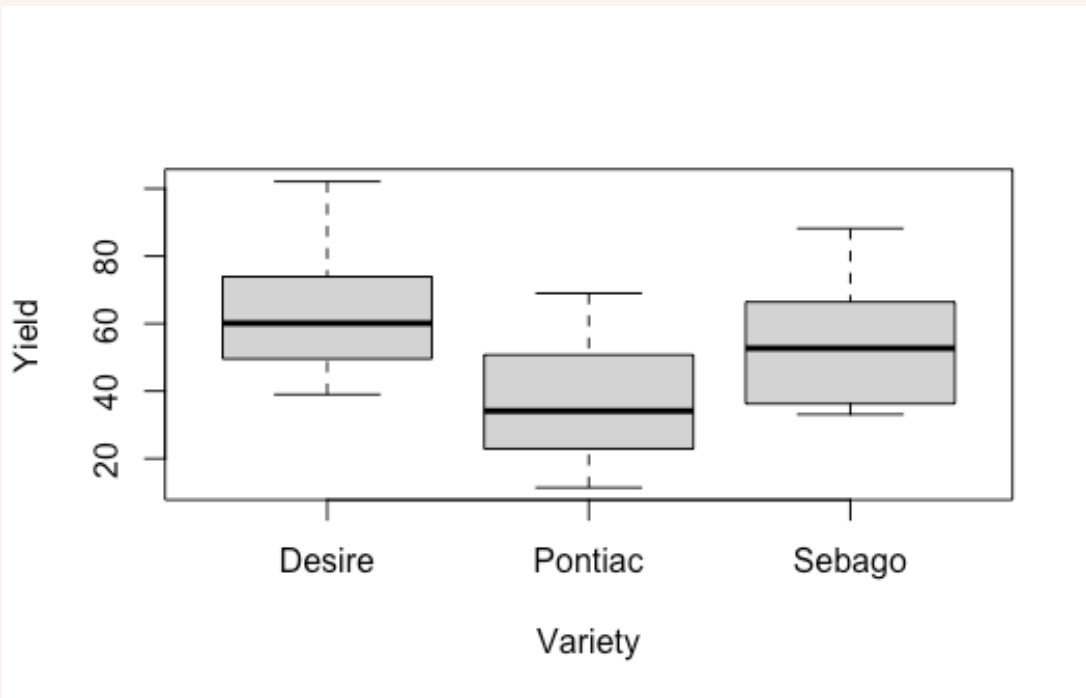
```
CODE
pot$Block<-as.factor(pot$Block)
pot$Variety<-as.factor(pot$Variety)
pot$Chemical<-as.factor(pot$Chemical)
str(pot)
```

```
OUTPUT
tibble [45 × 4] (S3: tbl_df/tbl/data.frame)
 $ Yield   : num [1:45] 34.1 76.3 72.2 51.9 42.9 62.1 66.3 39.1 75.1 41.7 ...
 $ Block   : Factor w/ 3 levels "1","2","3": 1 1 1 2 2 2 3 3 3 1 ...
 $ Variety : Factor w/ 3 levels "Desire","Pontiac",..: 2 3 3 1 3 2 3 3 1 2 ...
 $ Chemical: Factor w/ 5 levels "Control","CuOxy10",..: 3 4 5 3 2 4 4 2 5 5 ...
```

Graphical summaries for the main effects are easier to interpret than summary statistics.

The boxplots based on Variety show that Pontiac is the worst performing.

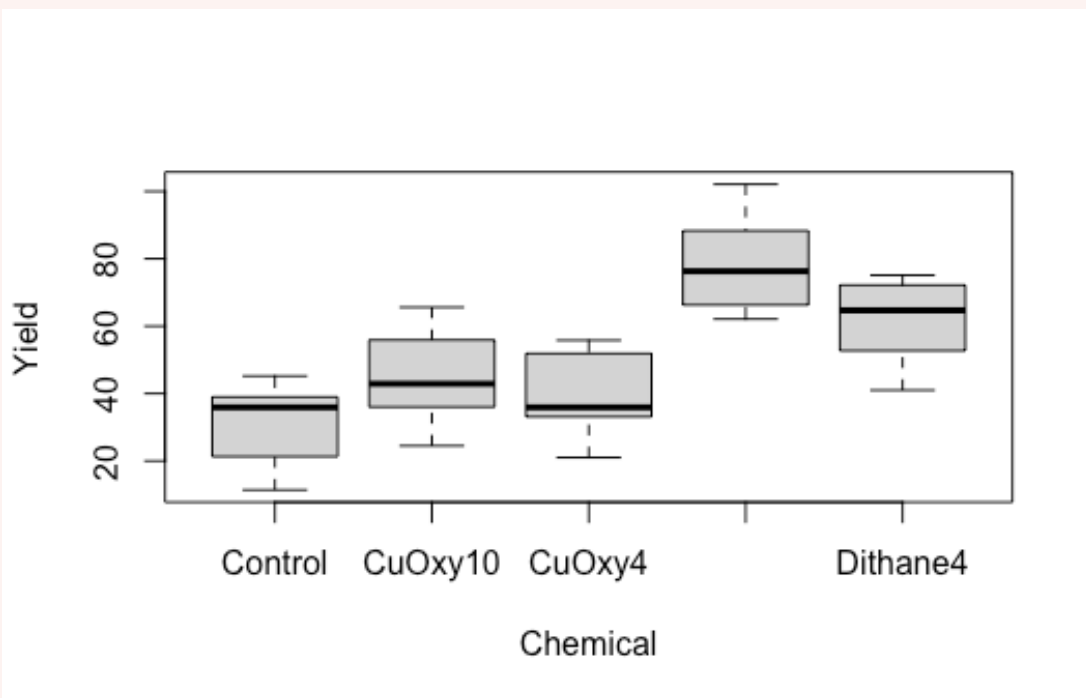
```
CODE
boxplot(Yield~Variety,data=pot,main=" ",xlab="Variety",ylab="Yield")
```



The boxplots for each level of Chemical show that Dithane10 results in the greatest yield.

CODE

```
boxplot(Yield~Chemical,data=pot,main=" ",xlab="Chemical",ylab="Yield")
```



The marginal means below show that applying no chemical (Control) to suppress potato blight results in the worse yields for all varieties. The combination of Desire and Dithane10 result in the greatest yield.

CODE

```
with(pot, tapply(Yield, list(Chemical=Chemical, Variety=Variety), mean) )
```

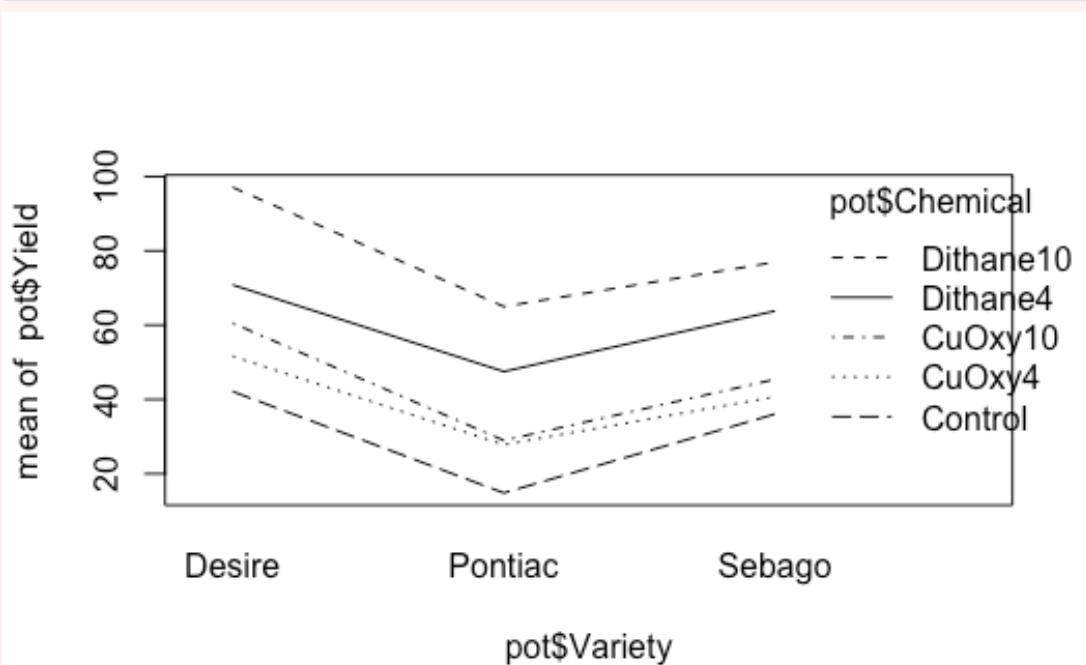
OUTPUT

	Variety		
Chemical	Desire	Pontiac	Sebago
Control	42.13333	14.83333	36.06667
CuOxy10	60.53333	28.96667	45.40000
CuOxy4	51.63333	27.80000	40.70000
Dithane10	97.10000	65.00000	76.93333
Dithane4	70.83333	47.50000	63.80000

The interaction plot below shows the lines to be reasonably parallel so an interaction effect is unlikely.

CODE

```
interaction.plot(pot$Variety, pot$Chemical, pot$Yield)
```



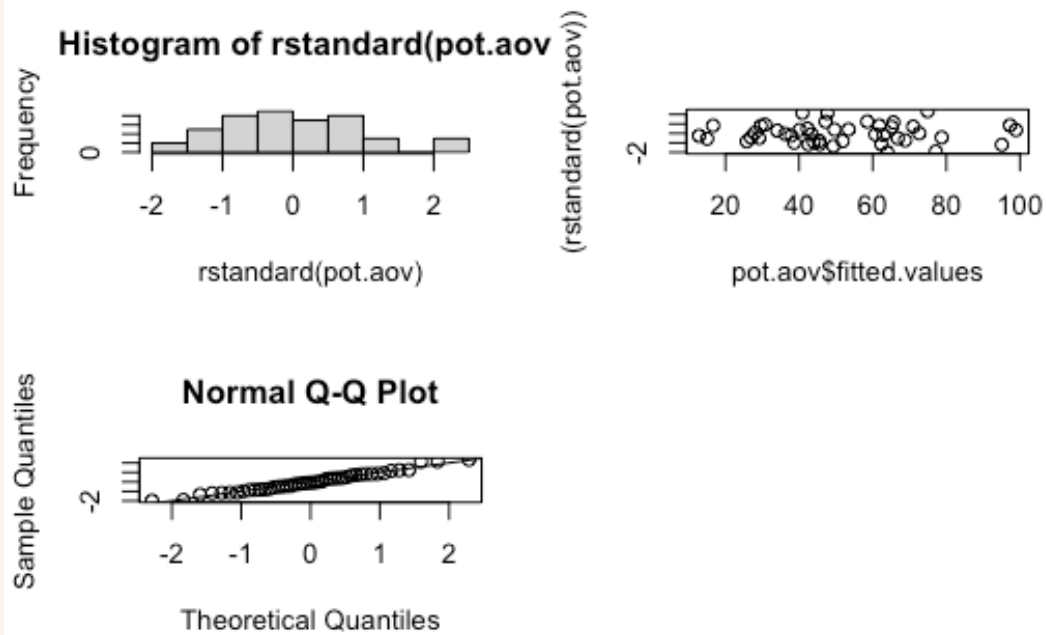
The residual diagnostics show that the data is normally distributed, has constant variance and has no obvious outliers.

CODE

```

pot.aov<-aov(Yield~Block+Chemical*Variety,data=pot)
par(mfrow=c(2,2))
hist(rstandard(pot.aov))
plot(pot.aov$fitted.values,(rstandard(pot.aov)))
qqnorm(rstandard(pot.aov))
abline(0,1)

```



or

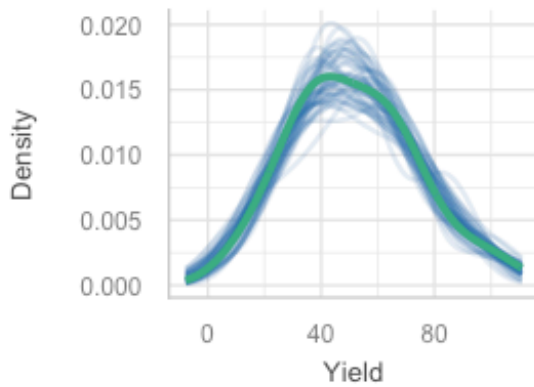
```

CODE
#library(performance)
check_model(pot.aov)

```

Posterior Predictive Check

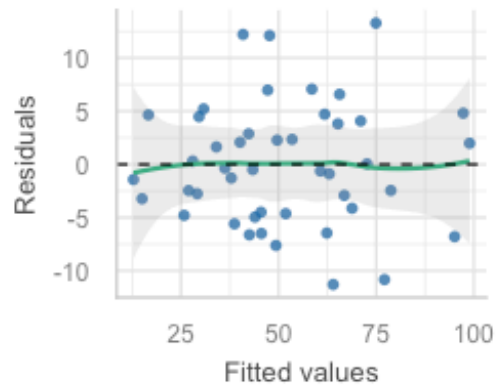
Model-predicted lines should resemble observed data



— Observed data — Model-predicted data

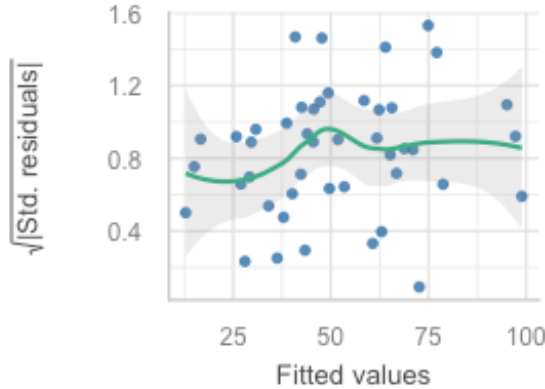
Linearity

Reference line should be flat and horizontal



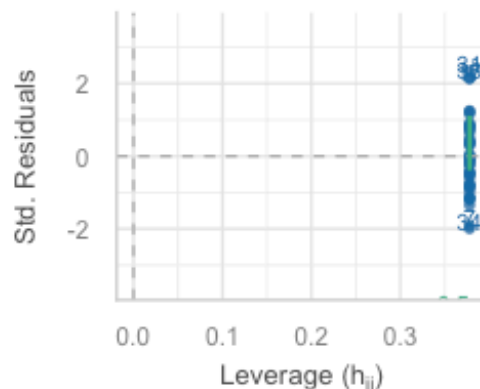
Homogeneity of Variance

Reference line should be flat and horizontal



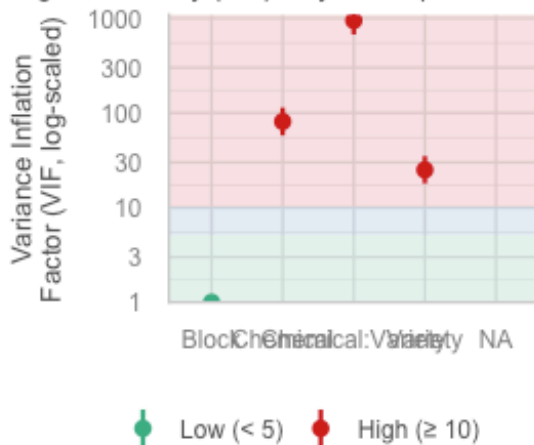
Influential Observations

Points should be inside the contour lines



Collinearity

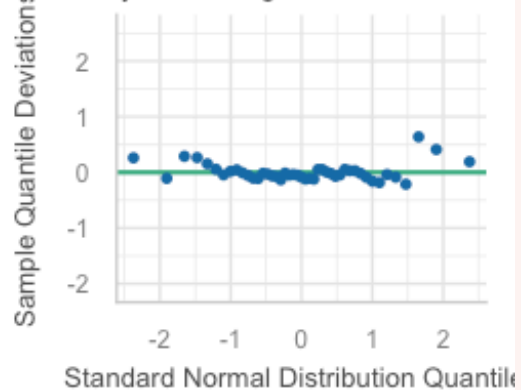
High collinearity (VIF) may inflate parameter estimates



● Low (< 5) ● High (≥ 10) NA

Normality of Residuals

Points should fall along the line



The ANOVA shows that interactions is not significant so we can examine the main effects individually. Both are significant with P-values less than 0.05.

```
CODE
summary(pot.aov)
```

```
OUTPUT

      Df Sum Sq Mean Sq F value    Pr(>F)
Block    2    111      55   1.071    0.356
Chemical  4 13251    3313  64.109 1.09e-13 ***
Variety   2   5762    2881  55.756 1.72e-10 ***
Chemical:Variety  8    259     32   0.628    0.748
Residuals 28   1447     52
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Tukey Post-hoc tests on the main effects is shown below.

```
CODE
emmeans(pot.aov, pairwise ~ Variety)
```

```
OUTPUT
NOTE: Results may be misleading due to involvement in interactions
```

```
OUTPUT
$emmeans
  Variety emmean   SE df lower.CL upper.CL
Desire    64.4 1.86 28    60.6    68.2
Pontiac    36.8 1.86 28    33.0    40.6
Sebago     52.6 1.86 28    48.8    56.4

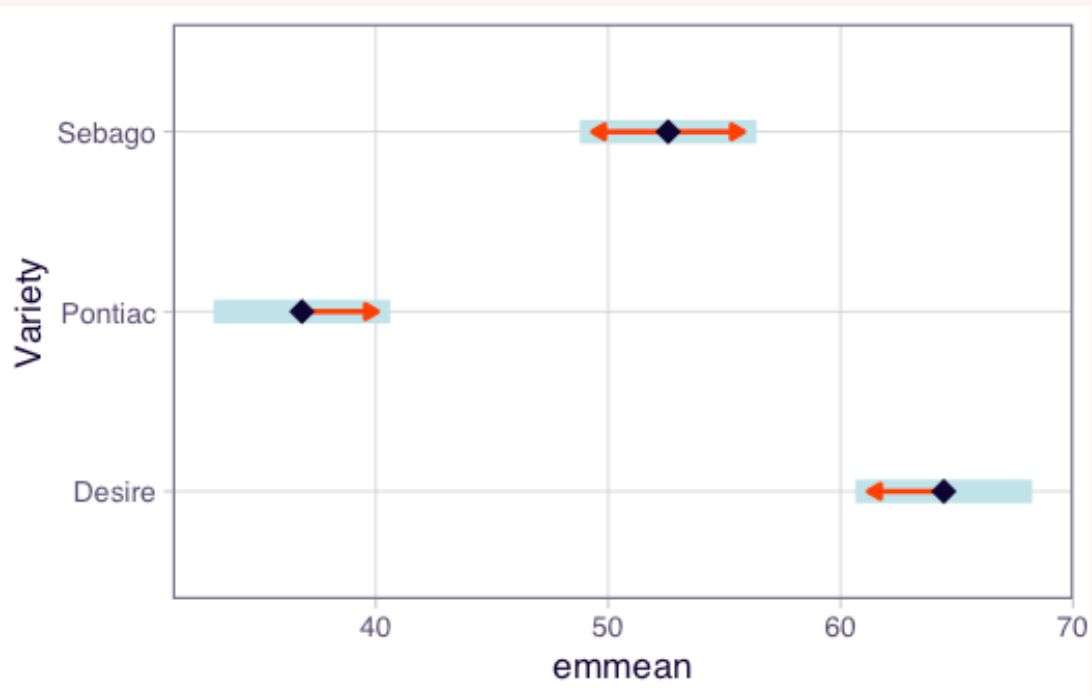
Results are averaged over the levels of: Block, Chemical
Confidence level used: 0.95

$contrasts
  contrast      estimate    SE df t.ratio p.value
Desire - Pontiac    27.6 2.62 28   10.525 <.0001
Desire - Sebago     11.9 2.62 28    4.521 0.0003
Pontiac - Sebago    -15.8 2.62 28   -6.004 <.0001

Results are averaged over the levels of: Block, Chemical
P value adjustment: tukey method for comparing a family of 3 estimates
```

```
CODE
plot(emmeans(pot.aov, pairwise ~ Variety), comparisons = TRUE)
```

```
OUTPUT
NOTE: Results may be misleading due to involvement in interactions
```



CODE

```
emmeans(pot.aov, pairwise ~ Chemical)
```

OUTPUT

NOTE: Results may be misleading due to involvement in interactions

OUTPUT

\$emmeans

Chemical	emmean	SE	df	lower.CL	upper.CL
Control	31.0	2.4	28	26.1	35.9
CuOxy10	45.0	2.4	28	40.1	49.9
CuOxy4	40.0	2.4	28	35.1	45.0
Dithane10	79.7	2.4	28	74.8	84.6
Dithane4	60.7	2.4	28	55.8	65.6

Results are averaged over the levels of: Block, Variety
Confidence level used: 0.95

\$contrasts

contrast	estimate	SE	df	t.ratio	p.value
Control - CuOxy10	-13.96	3.39	28	-4.118	0.0026
Control - CuOxy4	-9.03	3.39	28	-2.666	0.0852
Control - Dithane10	-48.67	3.39	28	-14.362	<.0001
Control - Dithane4	-29.70	3.39	28	-8.765	<.0001
CuOxy10 - CuOxy4	4.92	3.39	28	1.453	0.6003
CuOxy10 - Dithane10	-34.71	3.39	28	-10.243	<.0001
CuOxy10 - Dithane4	-15.74	3.39	28	-4.646	0.0006
CuOxy4 - Dithane10	-39.63	3.39	28	-11.696	<.0001
CuOxy4 - Dithane4	-20.67	3.39	28	-6.099	<.0001
Dithane10 - Dithane4	18.97	3.39	28	5.597	0.0001

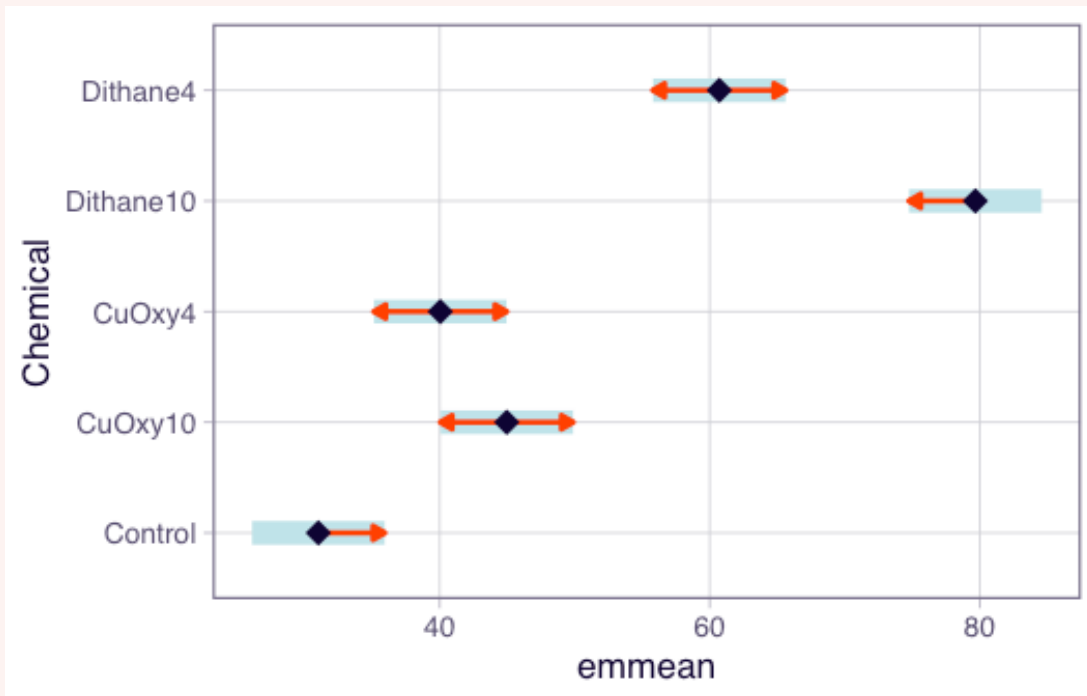
Results are averaged over the levels of: Block, Variety
P value adjustment: tukey method for comparing a family of 5 estimates

CODE

```
plot(emmeans(pot.aov, pairwise ~ Chemical), comparisons = TRUE)
```

OUTPUT

NOTE: Results may be misleading due to involvement in interactions



The results are quite clean with one level resulting in significantly greater mean yield for both treatment factors; Desire for Variety and Dithane10 for Chemical. From this we would recommend the use of Desire and Dithane10 for maximise yield via supressing potato blight.